

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)			Similarity – both have the same orbit time [of 24 hrs] (1) Difference – an object in geosynchronous orbit returns to the same point in the sky every 24 hours - an object in geostationary orbit always stays in the same position above the Earth - geostationary always orbits above the equator or converse (1)	2			2		
	(b)	(i)		[Satellite] 3		1		1		
		(ii)		$3 \times 10^8 = \frac{\text{distance}}{0.48}$ (1) substitution Distance = $0.48 \times 3 \times 10^8$ (1) rearrangement Distance = 1.44×10^8 [m] (1) answer	1	1 1		3	3	
		(iii)		$\frac{1.44 \times 10^8 \text{ ecf}}{36\,000\,000 \text{ ecf}} = 4$ (1) units must be consistent to award the mark So arrives at base station C (1) workings must be shown to award this mark If satellite 3 is not chosen in part (i) expect the following values – Satellite 1 = 6.71 award only 1 mark Satellite 2 = 6 (1) so arrives at A (1) Satellite 4 = 284 (1) so arrives at B (1) Satellite 5 = 132 (1) so arrives at A (1) Satellite 6 = 7.13 award only 1 mark		1 1		2	1	
				Question 6 total	3	5	0	8	4	0